
Channels to Targets: A Closed-Form Coordinate System for Silence-Blindness in LLM Observers

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Abstract

1 Introduction

An observer reads a transcript in which k of N analysts have filed a report, under a stated rule that a report is filed only on a positive finding. The $N - k$ analysts who filed nothing have not been silent about nothing. Under that rule each absence is a negative finding, and an observer who conditions only on what appears has discarded most of the record.

Our starting point is that this situation is small enough to solve exactly. Once the channels a transcript exposes are written down, the operations an observer must perform and the posterior it should hold both follow in closed form. An elicited answer is then not simply right or wrong: it has a location on an interval whose endpoints are named beliefs, and the distance and the direction are both measurable.

The design's asset is that the map from channels to targets is many to one: three transcripts requiring three different amounts of work return the same correct answer, so a comparison across them holds the target fixed and varies only what the observer must do to reach it. This is what lets us ask EIML's question about benchmarking blind spots in a form with an answer, since the blind spot has coordinates rather than a name.

- **An instrument.** A closed-form map from the channel set a transcript exposes to the operations required and the belief a correct observer should hold, with three reference beliefs and an exact pivot that makes three conditions share one target.
- **Localizing the direction of error.** Among the five configurations admitted by the competence gate, in the two conditions that do not enumerate the absences, under-weighting is the only direction of error observed, and the error anchors on the silence-blind belief rather than on the prior.
- **Separating identification from interpretation.** How much of a configuration's silence deficit is failure to notice the absences rather than failure to convert them is a per-configuration quantity the instrument measures, and enumerating the absences repairs the first without repairing the second.
- **The instrument's own limits, reported as results.** Two structural quantities are welded together by the geometry, one of our registered checks does not measure what it names, and our primary registered prediction did not hold.

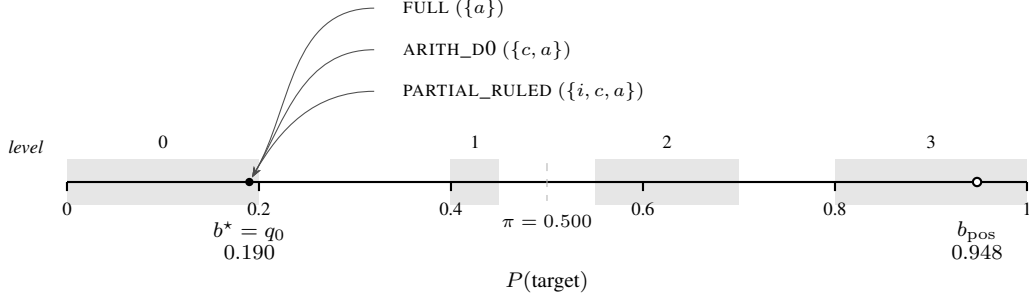


Figure 1: Three transcripts, one target. FULL, ARITH_D0, and PARTIAL_RULED display different channels and require different operations, but all three return the same correct belief, so the distance between an elicited answer and that belief is comparable across them.

2 The framework

Two candidates $A = \{a_0, a_1\}$ are under consideration and exactly one is responsible, with a uniform prior π . We take $|A| = 2$ throughout: the one-dimensionality that makes the word “coordinate” literal depends on it, and it is the source of a structural limitation we return to in Section 6. Each of the N analysts runs a binary assay returning POSITIVE or NEGATIVE. If a_0 is responsible an assay returns POSITIVE with probability p ; if a_1 is responsible, with probability $1 - p$; and the results are independent given which candidate is responsible. We take $p > 1/2$, so a positive is more consistent with a_0 than with a_1 and tilts belief toward a_0 . What the observer’s transcript shows of these outcomes is fixed by a *disclosure rule*, and it is this rule that our conditions vary: some display every result, others file a report only on a positive and leave the negatives to be recovered from N and the count of filed reports. In the dropout conditions, where only positives are filed, a positive additionally fails to be filed with probability δ . This is the bookbag-and-poker-chip paradigm of Phillips and Edwards [1966] conditioned on a disclosure rule. Across conditions the transcript therefore displays different channels, as Table 1 sets out, and it is the channel set a transcript exposes that determines both the operations a correct observer must perform and the belief it should hold.

Write $L(a)$ for the probability that a given analyst’s assay returns positive under candidate a , and let k be the number of positives among the N assays. Three beliefs are definable from the transcript alone:

$$b_{\text{pos}}(a) \propto L(a)^k, \quad (1)$$

$$q_{\delta}(a) \propto [(1 - \delta)L(a)]^k [1 - (1 - \delta)L(a)]^{N-k}, \quad (2)$$

$$q_0(a) \propto L(a)^k (1 - L(a))^{N-k}. \quad (3)$$

Equation (1) reads the positive filed reports as if they were the whole record. Equation (2) conditions on presences and absences alike, discounting each absence by the chance it reflects a dropout rather than a negative finding. Equation (3) treats every absence as certain disconfirmation.

We write $b^* := q_0$ at $\delta = 0$ only, where it is the correct posterior. At $\delta > 0$ the correct posterior is q_{δ} and q_0 becomes the *credulous point*, credulous in that it takes every absence at face value as a negative, lying past correct in the direction of over-reading, so draws below correct in the dropout conditions are readable rather than merely wrong.

Proposition 1 (The pivot). *At $\delta = 0$, $q_{\delta} = q_0 = b^*$.*

Proof. Setting $\delta = 0$ in (2) makes the discount factor $(1 - \delta)$ vanish from both exponentiated terms, so the silence factor $[1 - (1 - \delta)L(a)]^{N-k}$ becomes $[1 - L(a)]^{N-k}$, which is exactly the negative-result factor of (3). The two expressions are identical term by term. $\square$

The consequence is the design’s asset: conditions that print the absences, enumerate them, or leave them to be derived all return the same target, so their comparison is controlled. Every reference belief in this paper is computed over exact rationals, and Proposition 1 is asserted as exact rational equality across all fourteen cells rather than to a tolerance; no floating point appears anywhere in the model.

Table 1: The map. Each condition’s transcript exposes a different channel set, which determines both the operations a correct observer must perform and the belief it should hold. Rows 2 through 4 differ in operations and agree on target, by Proposition 1.

condition	channels in transcript	operations required	target
PARTIAL_BLIND	positives	aggregate	b_{pos}
FULL	positives, printed negatives	aggregate	b^*
ARITH_D0	positives, enumerated absences	convert, aggregate	b^*
PARTIAL RULED	positives, derivable absences	identify, convert, aggregate	b^*
ARITH_D30	positives, enumerated absences, δ	convert, aggregate, discount	q_δ
RULED_D30	positives, derivable absences, δ	identify, convert, aggregate, discount	q_δ

Lemma 1 (Ordering). Let $|A| = 2$ and $p > 1/2$, and consider a transcript with at least one absence ($N - k \geq 1$). For every $\delta \in (0, 1)$ the beliefs b_{pos} , q_δ , q_0 are strictly ordered in log-odds, and q_δ moves monotonically toward b_{pos} as δ increases.

Proof. Let a_0 be the candidate with $L(a_0) = p$ and a_1 the other, so $L(a_1) = 1 - p$ and $p > 1/2$ gives $L(a_0) > L(a_1)$. Write $\Lambda(b) := \log \frac{b(a_0)}{b(a_1)}$ for the log-odds a belief b places on a_0 . The three beliefs share the presence term $k \log \frac{p}{1-p}$ and differ only in how each of the $N - k$ absences is scored:

$$\begin{aligned} \Lambda(b_{\text{pos}}) &= k \log \frac{p}{1-p}, & \Lambda(q_\delta) &= k \log \frac{p}{1-p} + (N - k) g(\delta), \\ \Lambda(q_0) &= k \log \frac{p}{1-p} + (N - k) g(0), & g(\delta) &:= \log \frac{1 - (1 - \delta)p}{1 - (1 - \delta)(1 - p)}. \end{aligned}$$

Because $p > 1 - p$, the numerator of g is the smaller factor, so $g(\delta) < 0$ for all $\delta \in [0, 1]$: every absence strictly lowers the log-odds on a_0 . Moreover

$$g'(\delta) = \frac{p}{1 - (1 - \delta)p} - \frac{1 - p}{1 - (1 - \delta)(1 - p)} > 0,$$

since $x \mapsto x/(1 - (1 - \delta)x)$ is strictly increasing and $p > 1 - p$, so g climbs strictly from $g(0) = \log \frac{1-p}{p} < 0$ to $g(1) = 0$. Hence for any transcript with an absence ($N - k \geq 1$) and $0 < \delta < 1$,

$$\Lambda(q_0) < \Lambda(q_\delta) < \Lambda(b_{\text{pos}}),$$

the claimed strict ordering, and $\Lambda(q_\delta)$ rises monotonically to $\Lambda(b_{\text{pos}})$ as δ increases. The endpoints recover Proposition 1 at $\delta = 0$ and b_{pos} as $\delta \rightarrow 1$. $\square$

Four operations appear in Table 1. To *identify* is to determine which reports are absent; once N is stated this is a single subtraction, so noticing that reports are missing and naming which ones are missing are not separable stages. To *convert* is to read an absence as a negative finding. To *aggregate* is to combine the findings into a posterior. To *discount* is to weigh each absence against the stated dropout rate. The three conditions sharing a target require the strictly nested sets $\{a\} \subset \{c, a\} \subset \{i, c, a\}$, which is the ladder Example 2 walks. Two honest limits follow: aggregation is never identified in isolation, because no condition in our design sits below $\{a\}$, and discounting is identified only by the $\delta = 0.3$ conditions.

A three-way decomposition of total tracking error and a group-size scaling law follow from the same objects and are developed elsewhere.

3 The instrument

Answers are elicited as two enumerated fields, a candidate and a two-level confidence, giving four ordinal levels. The probability bands those levels occupy were measured rather than assumed, on both sides and without mirroring the low side from the high: a pre-test over 32 stimuli at three draws per stimulus across all seven configurations (659 of 672 usable) placed the switch edges at 0.20 and 0.80, with a low-high asymmetry of 0.000 at the widest for six of the seven configurations on the wording the main campaign uses. The single exception is *anchor*, at 0.1. Because the calibration

probe samples at a resolution of 0.10, the nominal cuts and the measured edges both fall inside the same sampled interval, so the invariance of level assignment across them is a fact about sampling resolution rather than a robustness result. Bands were never shown to the models.

Four constraints are enforced at build time. Every cell asserts that the silence-blind belief and the target fall in different levels, which holds fourteen for fourteen at $\delta = 0$ and nine for nine at $\delta = 0.3$. Prompts never use silence vocabulary, enforced by a forbidden-word list. Every stimulus appears in both label orderings. Count-control cells invert the correct level, so a configuration that always answers high-confidence is separable from a competent one.

The bands have real gaps between them, and the viability filter is how the design survives those gaps rather than ignoring them. Five of the fourteen cells fail it on gap placement: two put the dropout target at 0.366 and 0.392, inside the gap between bands 0 and 1, and three put it between 0.512 and 0.524, inside the gap that straddles the prior. That is what reduces fourteen cells to the nine carrying the dropout conditions. The filter is also stricter than cell validity requires, since a cell carrying those conditions needs its dropout target to be distinguishable from both the silence-blind belief and the credulous point, which is why stating a dropout rate can change the correct answer.

We report $s_1 = |\ell_{\text{obs}} - \ell_{\text{cor}}|$, the signed $s_2 = \ell_{\text{obs}} - \ell_{\text{cor}}$ with positive values meaning under-weighted toward the target, and $s_3 = P(\ell_{\text{obs}} = \ell_{\text{cor}})$, where ℓ_{cor} is the level of the target read off Table 1.

A configuration failing the matched printed-negatives condition is excluded, because a failure on the derivable-absences condition from a configuration that also fails the printed one is a floor effect rather than a silence result. Five of seven configurations are admitted, at mean s_1 of 0.200, 0.421, 0.457, 0.752 and 0.764 bins; the two excluded sit at 1.609 (deepseek) and 1.800 (anchor). The gap between 0.764 and 1.609 means membership does not depend on where in that interval the bar is placed.

The configuration the gate excludes at 1.800 is also the one that fails the pre-test’s reachability check, the only one whose switch point moves between wordings, and the only one with nonzero low-high asymmetry, so four instrument checks and one behavioural gate select the same configuration.

Across five experimental arms the campaign holds 7,051 usable draws of 7,210 attempted, at five draws per stimulus; the pre-test described above contributes a further 659 of 672 on a separate stimulus set at three draws per stimulus.

Predictions, gates, and kill conditions were fixed in a specification document and implemented as executable decision rules; Section 6 scores them and states plainly what our record does and does not establish about when they were fixed.

4 Example 1: which direction does the error run?

Whether language models update too little or too much is contested, and the two closest anchors disagree on populations that do not overlap. Imran et al. [2025] report that pre-trained-only checkpoints update less than Bayes requires, with the shortfall shrinking as models scale, measured over token log-probabilities; they describe the shortfall itself as unexplained and note that it may reflect how improbable their evidence strings are, so we take the direction from them and not a mechanism. Samanta et al. [2026] evaluate instruction-tuned models across turns and report both directions, with the tendency shifting across scale: smaller models stay near the middle of the interval, while larger ones can push toward its extremes when the evidence is skewed. Under-use relative to Bayes has been called conservatism since Edwards [1968]. The two accounts are hard to compare because each scores an elicited answer against a reference the same model supplied. Our target is computed from a stated disclosure rule over exact rationals, so no part of the correct answer depends on what probability a model assigns to any string, and both directions become positions on one axis rather than findings to adjudicate.

Both anchors were selected after our data existed: the campaign’s predictions were registered, the literature engagement was not.

The comparison uses two channel sets, printed negatives against derivable absences, which require different operations and share a target by Proposition 1. Because the target lies below the silence-blind belief in every cell, and lies in the bottom level in eleven of the fourteen, over-weighting is

structurally inexpressible outside the three cells where the target sits one level up. Those three cells carry every claim in this section, and no one-sidedness statement here can be pooled across the grid.

On those three cells the derivable-absences condition runs 47 to 0 toward under-weighting across the five admitted configurations, on 150 attempted draws of which 150 were usable. The competence gate selects on the printed-negatives condition and not on this one, so the result arm is unselected. At the registered unit of analysis, which pools mirrors and draws within a cell to a single modal level, eight of the fifteen configuration-by-cell units are displaced toward under-weighting and none toward over-weighting (two-sided exact sign test, $p = 0.008$); the remaining seven carry no displaced mass and are undefined rather than failed. The printed-negatives condition, which is the arm the gate selects on, runs 9 to 0 on the same 150 attempted draws, five of fifteen units displaced, $p = 0.063$. That arm is a control and not independent corroboration: every over-read it produces across the full roster comes from the two configurations the gate excluded. The direction persists under the easier channel set, and it does not need to reach significance there for the result to stand.

The error is not symmetric shrinkage toward the prior. In every configuration with error mass to test, the displaced mass sits above the prior or exactly at the level of the silence-blind belief, a region symmetric shrinkage cannot reach: 8 of 91 error draws, 50 of 80, 140 of 140 with 108 landing exactly at that level, and 17 of 52. The fifth admitted configuration produces no error draws at all, so the test is undefined there rather than failed. The test is well posed for a stateable reason: the uniform prior at 0.5 falls inside the band gap that straddles it exactly, so two levels lie wholly below it and two wholly above, with none ambiguous. Occupancy of the silence-blind level falls in none of the five and rises in four. An ordinal extremity account predicts the opposite of what we see: on a four-level scale the two middle levels are the ones an extremity effect would evacuate outward, so it predicts mass accumulating at the two endpoints. Of the 47 displaced draws, 44 land at the level of the silence-blind belief, which is not an endpoint, 3 land at the top level, and none at the bottom, so total endpoint mass is 3 in 150 attempted draws. The silence-blind account names an interior level in advance; extremity names the two levels that stay nearly empty. In all three configurations with displaced mass to test, mass at the silence-blind level outnumbers mass at the endpoint.

One-sidedness here is a property of these two conditions and not of the instrument. The over-direction is recoverable with the same models once the channel set changes, as Section 5 shows at both dropout rates, so the objection that our roster is too small to produce it becomes the observation that our instrument sees it only where the transcript enumerates the absences, which is the thesis.

This example required no new inference calls and runs entirely on draws collected before it was conceived.

5 Example 2: identification or interpretation?

When a system fails to use the absences, it may be failing to notice them or failing to convert them once noticed. In general these two are confounded, because a task that makes the absences easier to see usually also makes them easier to interpret.

Table 1 separates them without a new task. Its three shared-target conditions have strictly nested operation sets, so moving up the ladder deletes exactly one operation and changes nothing else, and the deletion is performed by the transcript rather than by an instruction. This does not have to be designed in; it falls out of the map.

The added rung contributes 980 calls, 959 usable, over a stimulus set fixed at a content hash; Section 6 states what that hash does and does not establish.

Deleting the identification step frees the answers. Residence at the silence-blind belief collapses in every non-ceiling admitted configuration, from 4.6 to 0.0, from 30.6 to 12.9, and from 77.1 to 50.0 percent of draws: the absences were going unnoticed. But the freed answers do not arrive at the target. Interior mass, the mass strictly between the silence-blind belief and the correct one, rises in every one of them, from 65.4 to 66.9, 28.4 to 42.4, and 17.1 to 27.9. Identification is repaired and conversion is not. One further configuration, flagged at ceiling by the registered guard and therefore outside this population, moves the other way, from 6.4 to 10.7, and we report it.

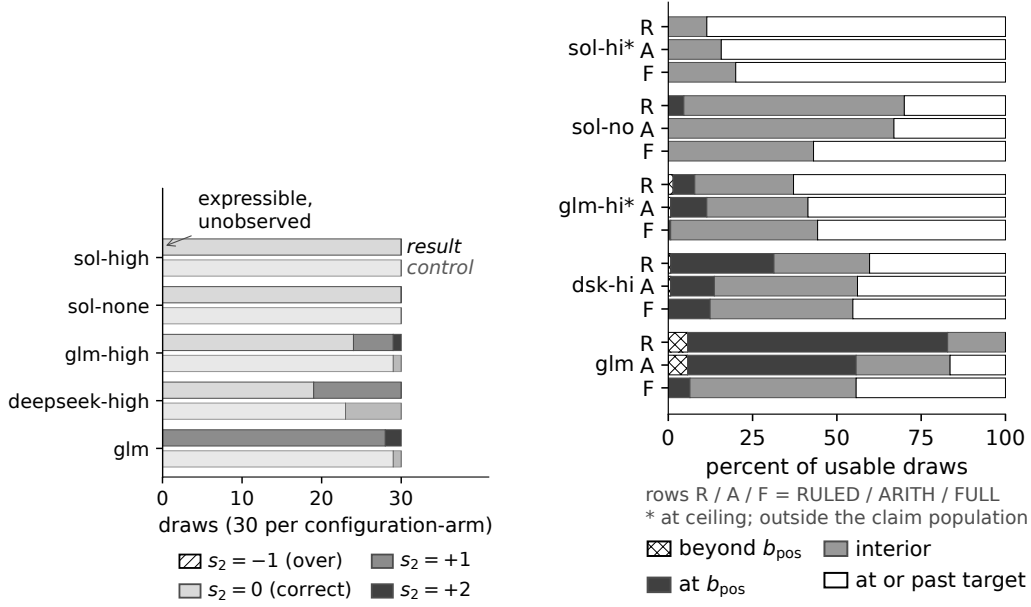


Figure 2: **(a)** Signed displacement on the three cells where over-weighting is expressible, for the five admitted configurations. The derivable-absences condition is the result arm; the printed-negatives condition is the control the competence gate selects on. The shaded region below zero is expressible and unobserved. **(b)** The ladder: deleting one operation at a time along the nesting $\{i, c, a\} \supset \{c, a\} \supset \{a\}$, with the distance to the silence-blind belief marked for reference. The drop and the residual gap are both visible.

The split is a per-configuration quantity and not a constant. All five admitted configurations receive a well-defined verdict: three numeric positions and two adjudicated at ceiling by the registered guard. Writing λ for the share of the gap between the two conditions that enumeration leaves unclosed, the closures span 28 to 114 percent of that gap, with the top of the range falling past the printed-negatives floor. The registration for this rung carried no numeric bounds, so we report λ as a measured descriptive quantity rather than scoring it against a prediction.

The same handout repairs no one once a dropout rate is stated. At $\delta = 0.3$ no configuration recovers more than 12.5 percent of its deficit, the admitted five recover at most 8.0 percent, and a worst-case reassignment of unusable draws moves the bound by no more than 0.20 bins. That interval contains all seven configurations and needs no gating argument. It is also where the geometry does work: the target moves to the discounted belief and the undiscounted one becomes a credulous point past correct, so draws below correct are readable as over-reading rather than as noise. One contrast we had fixed at a checkpoint failed on this arm and is retired rather than rescued.

The obvious alternative is that enumeration changes the record's length rather than its content, and there is real evidence for it: the count-control concordance fails for three configurations, and both ceiling configurations move the wrong way under enumeration, by 0.043 and 0.114 bins, which is the shape a length cost would take. Two things stand against it. Length cannot account for a gain of 0.493 bins, and a per-line effect fails to appear anywhere, ranging from -0.89 to $+0.17$ after a collinearity audit bounding the relevant correlation at 0.34. The concordance test also inherits the defect we describe in Section 6: its constancy criterion is confounded in the count-control group and clean in the group carrying this example's claims.

The two halves meet here. Once the absences are enumerated, over-reading appears without any dropout rate at all: two over-reads in 150 expressible admitted draws on the enumerated condition, against none in 300 on the two conditions that do not enumerate, and the two configurations producing them are exactly the two with the largest repairs. The direction of error flips with the channel set, at both dropout rates.

Table 2: Registration scoring across three tiers. Tier labels record where an item was registered, not that its ordering against the campaign was verified; see Section 6.

item	tier and where registered	outcome
P3 (primary): gap > 0.5 in ≥ 2 of 3 families	Tier 1, specification, internal timestamp	Falsified under a strict reading of our own P5 exclusion clause (0 of 3); held in 1 of 3 under a loose one. The single family that clears it at 0.821 also fails P5. Both readings reported.
P1 and P2, the two halves	Tier 1, same	1 of 3 each, in <i>different</i> families. No family carries both halves.
Non-control cell bar	Tier 1, amended	Specification text says 8 of 11; executed as 6 of 8. The non-control set has eight members, so the original figure was a miscount. Proportion $0.727 \rightarrow 0.75$. Dated amendment.
Enumerated-rung λ pre- dictions	Tier 2, content hash	Directional, no numeric bounds. Scored as directional.
Enumerated-rung ceiling guard	Tier 2, same	Sharp 0.15-bin threshold, scored cleanly.
Cross-regime contrast	Tier 3, fixed at a check- point, absent from the specification	Did not hold. Retired rather than rescued.

This recovers, by a different method, the failure Min et al. [2026] report as over-closure: treating missing support as settled is the closed-world assumption applied where it does not hold, which in our coordinates is the credulous point. They ask whether an absence licenses a negative answer at all, holding the question fixed and varying how coverage is expressed; we assume licensing and ask how far the belief should move, holding the target fixed and varying which operations the transcript requires.

6 What the instrument cannot identify

Our specification defined a three-band outcome space in advance: a pass required the effect in at least two of three families, zero families counted as falsification with the registered consequence that this becomes a negative-results report, and one family was neither. Naming the bands is what makes the result below reportable rather than merely disappointing. One thing our record cannot do is order itself. Nothing in the repository establishes that any registration predates the calls it registers, for the specification, the stimulus set, or the band pre-test alike; version control begins after both campaigns, draw records carry no wall-clock field, and the specification file was edited after the campaign it registers. What is verifiable is content integrity and deterministic regeneration, and we describe every decision rule as fixed at a checkpoint before its test ran, which is a claim about our process and not about verified ordering.

The instrument’s limits are not a list of unrelated caveats. Three of them are the same shape: a pair of properties one would want to vary independently, fused by the geometry of the construction.

First, count and weight are anti-correlated by construction: holding the target fixed while raising the number of absences from four to twelve forces the assay sensitivity down from 0.90 to 0.56, and the evidence the silences carry falls from 5.61 through 1.83 to 0.71 nats, so tripling the count drops the evidence roughly eightfold. This is closed form and needs no draws. Second, over-reading is testable only in the weak-evidence corner, and the coupling is forced: the three cells where it is expressible carry silence evidence around fifty times weaker than the strongest cells in the grid, because over-reading needs headroom below the target, headroom requires the target to be high, and a high target means the silences carried little. Third, the response scale makes over-reads harder to see than under-reads: in those same cells the target sits inside the narrowest band in the instrument, with a gap of 0.20 below it and 0.10 above, so registering an over-read requires roughly twice the excursion in probability that registering an under-read does. That asymmetry ships with the zero reported in Section 4.

A fourth limit is our own. One registered check requires the modal answer to stay constant as the report count varies, but in the count-control group the silence-blind belief is pinned while the correct

level moves, so a configuration tracking the target fails the check and one parked at the silence-blind belief passes it. The most silence-blind configuration in our roster returns a perfectly constant row. This is why Table 2 reports the primary outcome under both readings of that check’s exclusion clause.

A fifth is that our normalised statistic is span-dependent, since the spans available across the control sweep differ and cannot be pooled; we retire it and report it rather than dropping it quietly. Its companion is structural and is a fourth instance of the pattern above. With two candidates the silence-blind belief is at least 0.5 by construction, so it occupies only two of the four bins and the falsification test is always a one-bin discrimination; in the cells carrying Section 4’s result the intermediate region is empty outright, so the fair symmetry test and the instrument’s narrowest limitation are the same geometry. The repair is stated with the limitation: both follow from fixing the candidate set at two, an axis we dropped deliberately and mark for revisit.

A sixth is that the dropout conditions are mechanically easier under an ordinal statistic and cannot serve as the harder test they appear to be. Relatedly, the one-sidedness reported in Section 4 belongs to the two conditions that do not enumerate the absences and not to the no-dropout regime as such: enumeration produces over-reading with no dropout at all, and under stated dropout 6.3 and 14.6 percent of draws across all seven configurations sit below correct.

7 Related work and limitations

Our silence-blind belief trims the record where the correct update censors it, in the sense of de Punder et al. [2026]: the omitted mass is exactly the silence probability, and that a divergence restricted to a sub-event loses its non-negativity is established there and in de Punder et al. [2024] rather than here. The same distinction appears in the knowledge-base literature as the gap between closed-world and open-world readings of a missing fact [Razniewski et al., 2024]. Our contribution is the instrument built on that distinction, not the distinction.

That language models mishandle absent evidence is known [Min et al., 2026, Fu et al., 2025]. Those results ask whether an absence licenses an inference at all; we ask how far a belief should move once it does, and only the second question requires a target computed from the disclosure rule.

Belief dynamics in multi-agent settings already admit martingale and Lyapunov treatments [Choi et al., 2025]. The dynamical frame is not our contribution here. Our setting is single-round and static by choice, because holding the target fixed is what makes the channel-set manipulation a controlled comparison.

Three limitations bound what we claim: the roster is seven open-weight configurations, comparable in breadth to the rosters our two closest anchors use; the candidate set is fixed at two, which is what forces the one-bin degeneracy described in Section 6; and the setting is single-round and static.

Reproduction. Stimuli, decision rules, draw records, and analysis scripts are available at commit 9ade4d4, together with a provenance statement describing exactly what the record establishes and what it does not.

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